

# EN4730 Convex Engineering Design Assignment 01

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1. Determine the following functions and sets are convex, concave or neither convex nor concave with a justification.

(a)  $f(x) = -x \log(x)$ .

(b)  $f(\mathbf{x}) = e^{x_1} + e^{x_2} + x_1^2$ .

(c)  $f(\mathbf{x}) = 2x_1^2 - 6x_1x_2 + 2x_2^2$ .

(d)  $f(\mathbf{x}) = \frac{1}{2}(ax_1^2 + bx_2^2)$  with  $a, b \leq 0$ .

(e)  $f(\mathbf{x}) = \frac{1}{2}(ax_1^2 + bx_2^2)$  with  $a \times b > 0$ .

(f)  $f(x) = \frac{x^\beta - 2}{\beta}$ .

(g)  $S = \{(x_1, x_2) \in \mathbb{R}^2 \mid x_1 \geq 0, x_2 \geq 0, 2x_1 + x_2 \geq 2, x_1 + 2x_2 \leq 6\}$ .

(h)  $S = \{(x_1, x_2) \in \mathbb{R}^2 \mid x_1 \geq 0, x_2 \geq 0, 2x_1^2 + x_2 \geq 2, x_1 + 2x_2 \leq 4\}$ .

(i)  $S = \{(x_1, x_2) \in \mathbb{R}^2 \mid x_1 \geq 0, x_2 \geq 0, x_1^2 + x_2^2 < 16, x_1 + 2x_2 < 4\}$ .

(j)  $S = \{(x_1, x_2, a) \in \mathbb{R}^3 \mid x_1 \geq 0, x_2 \geq 0, x_1^2 + x_2^2 = a^2\}$ .

[15 marks]

2. Consider the following constrained optimization problem

$$\begin{aligned} & \text{minimize } \|\mathbf{x}\|_1, \\ & \text{subject to } \mathbf{x}^T \mathbf{a} = c. \end{aligned}$$

where  $\mathbf{x} = [x_1 \ x_2]^T$ ,  $\mathbf{a} = [a_1 \ a_2]^T \in \mathbb{R}^n$ , and  $c \in \mathbb{R}$ .

- (a) Plot level curves (contour plots) for

i.  $\|\mathbf{x}\|_1 = 4$

ii.  $\|\mathbf{x}\|_1 = 8$

iii.  $\|\mathbf{x}\|_1 = 12$

with  $\mathbf{x} \in \mathbb{R}^2$ .

[5 marks]

- (b) Determine the condition on  $a_1$  and  $a_2$  under which the problem has a **unique solution**. Justify your answer geometrically. [12 marks]

- (c) Determine the condition on  $a_1$  and  $a_2$  under which the problem has **infinitely many solutions**. Justify your answer geometrically. [12 marks]

- (d) Provide one explicit example of  $(a_1, a_2)$  for each case above. [6 marks]

3. Consider the following constrained optimization problem

$$\begin{array}{ll}\text{minimize} & f(\mathbf{x}), \\ \text{subject to} & \mathbf{Ax} \leq \mathbf{c}.\end{array}$$

where  $\mathbf{x} \in \mathbb{R}^n$ ,  $\mathbf{A} \in \mathbb{R}^{m \times n}$  and  $\mathbf{c} \in \mathbb{R}^m$ . It is given that

$$\mathbf{A} = \begin{bmatrix} -1 & 0 \\ 1 & 0 \\ 0 & -1 \\ 1 & 2 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} 0 \\ 0 \\ 4 \\ 5 \end{bmatrix}.$$

(a) Draw feasible region of  $\mathbf{Ax} \leq \mathbf{c}$ .

**[10 marks]**

(b) Write down a function  $f(\mathbf{x})$  that correspond to each of the following cases:

- i. One unique solution,
- ii. One unique solution with either  $x_1 = 0$  or  $x_2 = 0$ .

**[40 marks]**

## **Submission**

- Upload a report as a pdf file named as "yourindexno\_EN4730\_A01.pdf". Include the index number and the name within the report as well.
- The interpretation of results and the discussion are important in the report.
- Pay careful attention to formatting such as font size, spacing, and margins.
- Include a title page with necessary information (e.g., title, author, date, index no).
- Use consistent and professional formatting throughout the document.
- Plagiarism will be checked and in cases of plagiarism, an extra penalty of 50% will be applied. In case of copying from each other, both parties involved will receive a grade of zero for the assignment. Academic integrity is of utmost importance, and any form of plagiarism<sup>1</sup> or cheating will not be tolerated.
- An extra penalty of 20% is applied for late submission.

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<sup>1</sup><https://en.wikipedia.org/wiki/Plagiarism>